

Hydro Pacifique: Local Database Guide

Purpose: a simple explanation of the demonstration data and how the application connects it.

1. Local setup

The application uses the local MySQL database named `hp-data-fj`. The PHP platform configuration connects to MySQL on 127.0.0.1 and selects the Pacific territory. The web server runs at `http://127.0.0.1:8000`.

2. Login user

The local administrator account is stored in `ad_user`.

Username: `anuj`

Password: `Anuj@2026!`

User ID: 86

Active: yes; administrator: yes; language: English

The permissions for this user are stored in `ad_user_acces`. User 86 has data, expert data, parameter, and configuration access. The account can therefore open protected application pages.

3. Territory, owner, and reference data

The platform row is in `geo_territoire`: territory ID 6, initialization code `Pacific`, language `en`, timezone `Pacific/Tahiti`.

The Data Owner filter reads `ad_service`, not the text field `station.proprio_station`. Service ID 1 is named `Hydro Pacifique`. A station points to this service through `station.id_service = 1`.

The equipment type is stored in `eq_type`. Type ID 11 is an active surface-water type. The station's `station_type = 11` uses this reference. The water-level time-series type is stored in `data_type`: type ID 65, code `WLR`, unit `mm`.

4. River water-level station

The original demonstration station is in `station`:

ID 2575

Name: Nuku'alofa River Water Level

Code: TGA-NKL-WLR

Coordinates: latitude -21.1394, longitude -175.2048

River: Nuku'alofa River, linked by `station.id_riviere` to `geo_riviere.id`

Region: Nuku'alofa, Tonga, linked by `station.id_region` to `geo_region.id_region`

The river is stored in `geo_riviere`. The station is visible on the map only when it has valid coordinates. It is visible in the station dropdown because the dropdown joins `station` to `geo_region` and applies active, owner, and type filters.

The 720 hourly water-level readings are stored in `data_all`. Each reading has a timestamp, a value, and an `id_meta`. The related metadata row is in `data_meta`, linking the series to station 2575 and data type 65.

Relationship:

`station 2575 -> data_meta 88215 -> data_all readings`

5. Stream-gauging station

A second station was created for manual stream gauging:

ID 2576

Name: Nuku'alofa River Gauging Station

Code: TGA-NKL-JGE

Type: 11, active, continuous

Same Nuku'alofa coordinates and river as the water-level station

The main gauging records are in `data_jge`. There are three synthetic records:
2026-08-20: discharge 4.60

2026-08-21: discharge 5.10
2026-08-22: discharge 5.80

Each record belongs to station 2576 through `data_jge.id_station`. Detailed cross-section information is stored in `data_jge_bras`, and measurement verticals/points are stored in `data_jge_points`. One arm with three points was added.

Relationship:

`station 2576 -> data_jge -> data_jge_bras -> data_jge_points`

The current-meter reference records are in `eq_moulinet`. Three synthetic meters were added so the JGE edit page can display and sort its equipment list: C2 DEMO-001, C31 DEMO-002, and MOL DEMO-003.

6. Rating curve / ETL

The rating-curve page `data_etl.php` lists stations that have rows in `data_etl`. A curve was added for station 2576:

ETL ID: 1600

Period: 2026-08-20 through 2026-08-22

Its five height-to-discharge points are stored in `data_etl_data`:

height 0.50 -> discharge 1.20

height 0.70 -> discharge 2.80

height 0.90 -> discharge 5.80

height 1.10 -> discharge 9.60

height 1.30 -> discharge 14.50

Relationship:

`station 2576 -> data_etl 1600 -> data_etl_data points`

The page defaults to stations with a rating curve. Without a `data_etl` row, it correctly shows no station.

7. Groundwater well and diagnostic profile

The groundwater page `data_diag_piezo.php` only accepts piezometric stations with `station_type = 5` and at least one diagnostic profile. A separate well was created:

ID 2577

Name: Nuku'alofa Groundwater Well

Code: TGA-NKL-GW01

Coordinates: latitude -21.1394, longitude -175.2048

Well depth: 18.0 m; water-table depth: 4.2 m

The header reading is in `data_ra` with record ID 57654. It has conductivity 620.0 and temperature 25.1.

The five depth-profile rows are in `data_ra_piezo_profile`:

depth 0.0 m -> conductivity 410.0, temperature 24.8

depth 2.0 m -> conductivity 480.0, temperature 24.9

depth 4.0 m -> conductivity 590.0, temperature 25.0

depth 8.0 m -> conductivity 620.0, temperature 25.1

depth 12.0 m -> conductivity 635.0, temperature 25.2

Relationship:

`station 2577 (type 5) -> data_ra 57654 -> data_ra_piezo_profile rows`

8. Why filters sometimes hide a station

The shared filter file builds SQL conditions from saved values in `ad_user_filter`. Important conditions include:

Data Owner: `station.id_service` must match a selected `ad_service` ID.

Data type: `station.station_type` must match the selected type.

Active status: `station.active_station` must be 1 by default.

Station dropdown: station must have a matching `geo_region` row because the query uses an inner JOIN.

ETL page: station must have at least one `data_etl` row by default.

Groundwater diagnostics: `station_type` must be 5 and `data_ra` must have linked profile rows.

A saved station selection can also conflict with a saved type selection. For example, selecting the groundwater station while the type filter is still 11 returns no matching row. The filter code now ignores that incompatible selection instead of reading a null result.

9. Database backup

A SQL backup of the database is stored in the `db_backup` directory. The backup contains the database structure and the demonstration records. Take another backup before making large data changes.

10. Mental model

Think of `station` as the central object. Geographic tables describe where it is. `ad_service` and `eq_type` describe ownership and station category. Measurement tables attach observations to the station: `data_all` for time series, `data_jg` for manual gaugings, `data_etl` for rating curves, and `data_ra` plus `data_ra_piezo_profile` for groundwater diagnostics. The web pages are mostly different filtered views over these relationships.